Spatial Structure Facilitates Category Learning through Structured Representations

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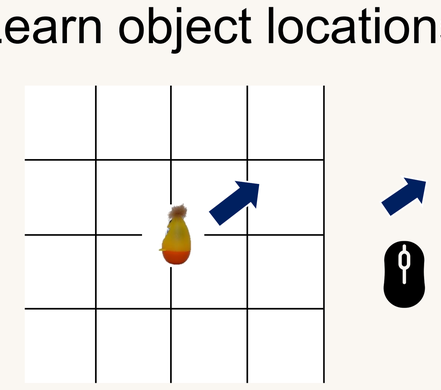
Why Should Space Matter?

- Spatial frameworks are thought to scaffold memory, inference, and abstract reasoning[1–4].
- But **does spatial structure transfer to non-spatial category learning, and if so, how?**

We asked:
1. Does structured spatial learning facilitate subsequent non-spatial category learning?
2. Can the geometry of learned representations explain this behavioral advantage?

Paradigm

Learn object locations

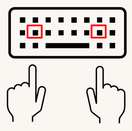


(Remove the spatial context)

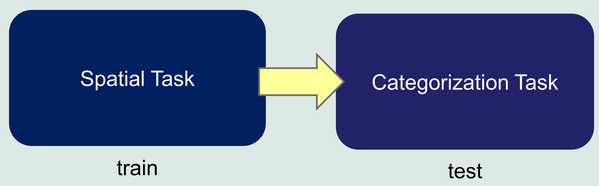
Categorize the same objects



"Press 'A' for Brispo
Press 'L' for Daplen"



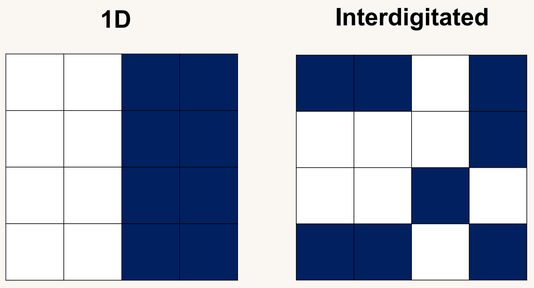
This sequence is repeated across **4 phases**, each with a new set of objects:



Metrics: accuracy, response time (RT)

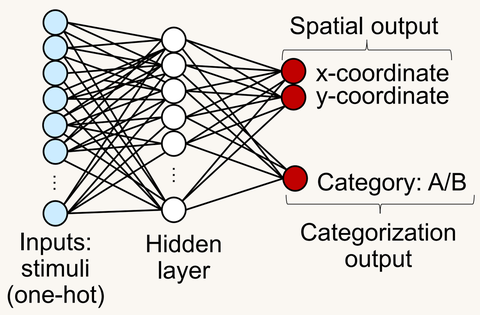
Shading indicates later category membership (not shown during spatial learning): grouped by side in 1D and intermixed in Interdigitated.

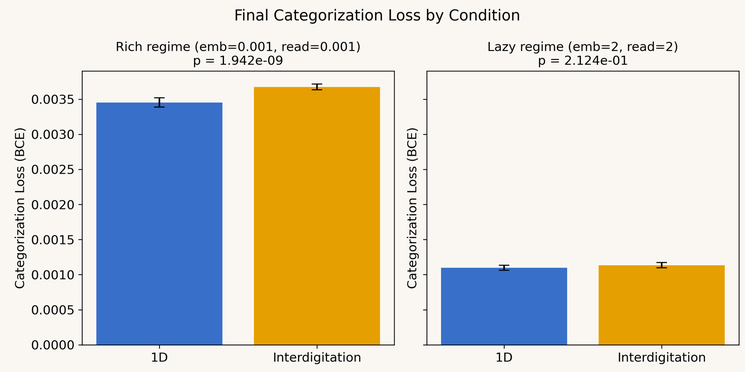




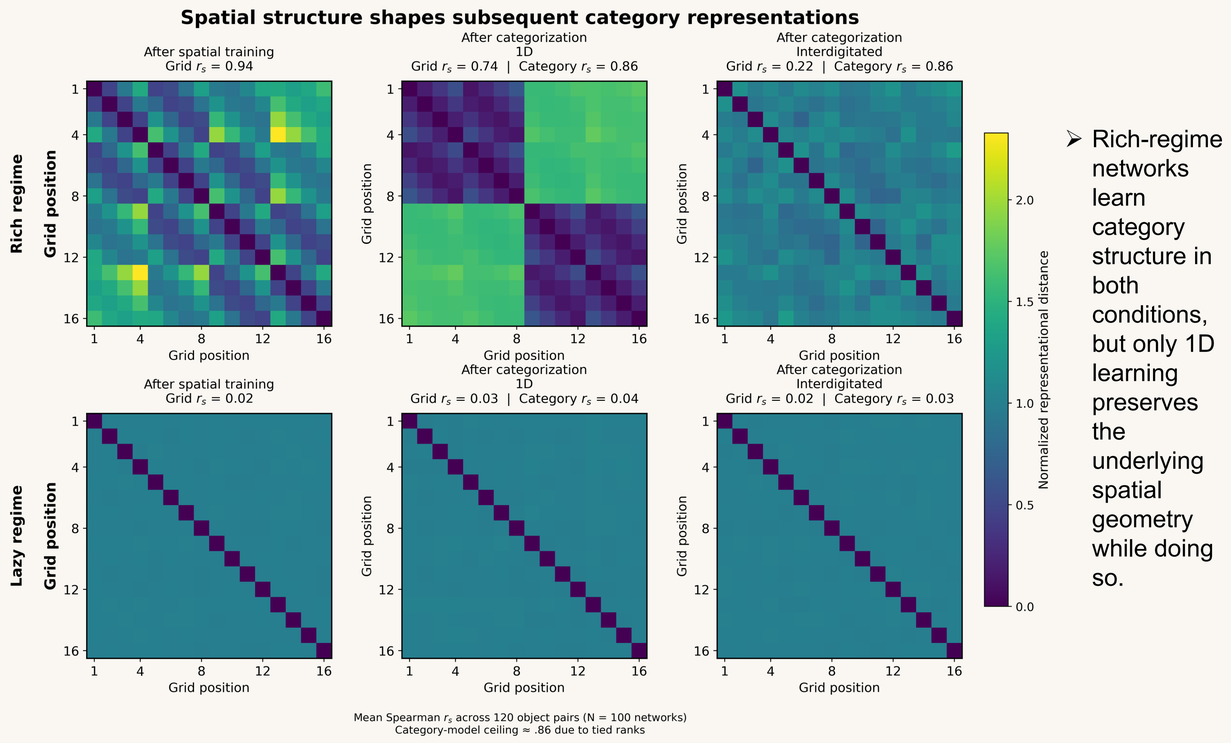
Computational Model

Linear Dual-Head Multi-layer Perceptron (MLP):





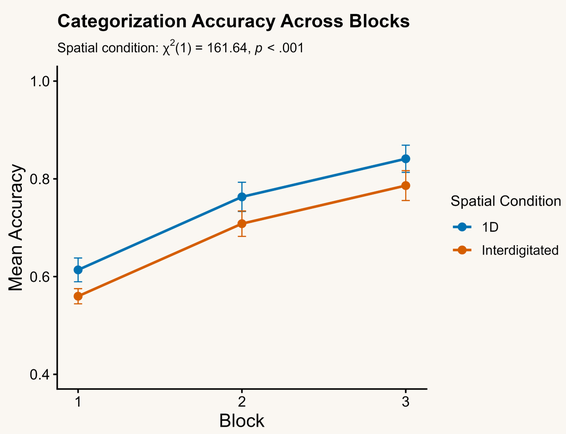
➢ To test whether representational change underlies this transfer, we trained a dual-head MLP on both the spatial task and the categorization task under two learning regimes: **rich** (representations reorganize during learning) and **lazy** (representations stay close to initialization). The same regime governs learning in both phases for a given model.

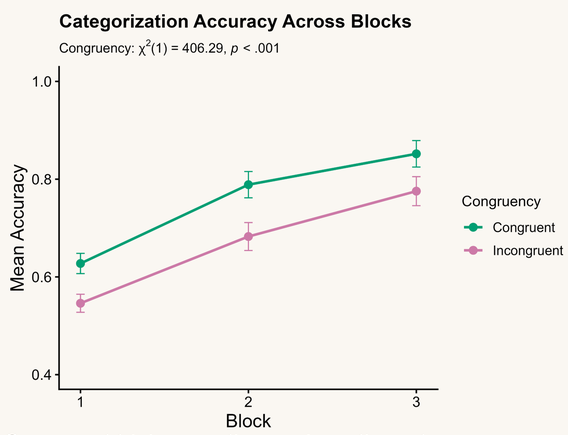


➢ Rich-regime networks learn category structure in both conditions, but only 1D learning preserves the underlying spatial geometry while doing so.

Human Behavior

N=100, within-subjects design





Congruent trials were those where the category response mapping matched the object's original spatial location

➢ Accuracy was higher following 1D than interdigitated learning, and for congruent than incongruent responses, with no clear interaction between the two.

Discussion

Spatial structure transferred beyond the spatial task: categorization was more accurate when the 1D structure of a category mapped onto the spatial layout learned earlier, compared to an interdigitated mapping.
Structure and response compatibility made separable contributions: congruent responses were more accurate than incongruent ones, but this did not appear to depend on spatial condition.
Representational change provides a plausible mechanism: the model reproduced the 1D advantage under rich learning, when spatial training reorganized its internal representations, but not under lazy learning.

Ongoing work: We are testing whether the same representational reorganization emerges in neural data during human learning.

References

1. Behrens, T. E. J., et al. (2018). What is a cognitive map? *Neuron, 100*, 490–509.
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4. Pesnot Lerousseau, J., & Summerfield, C. (2024). Space as a scaffold for rotational generalisation of abstract concepts. *eLife, 13*, RP93636.

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